

# Visualizations

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## Plot 1: Timeseries

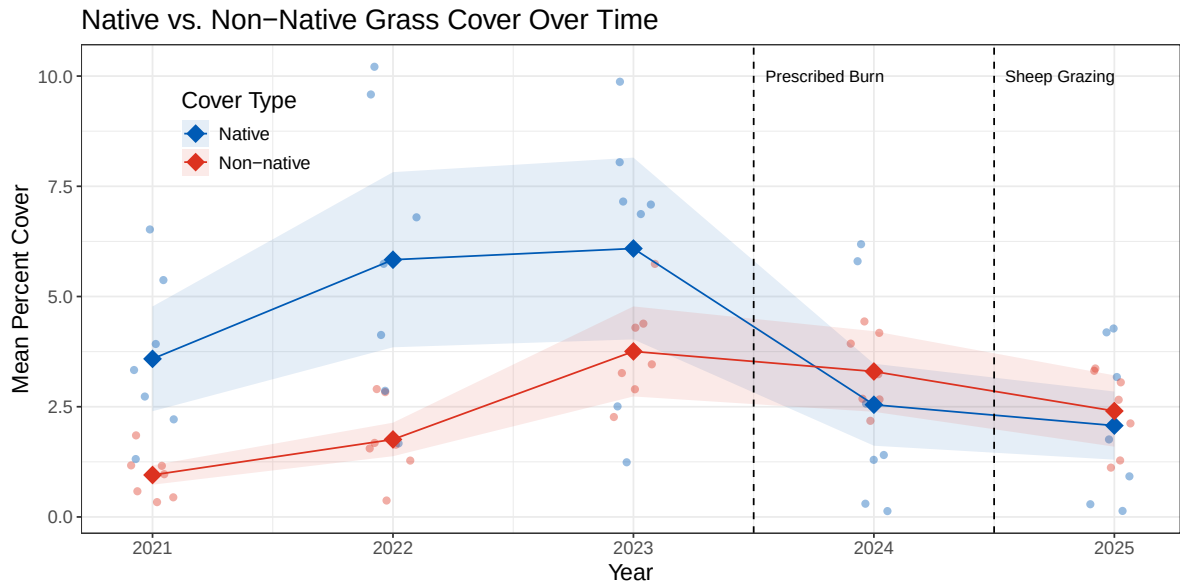


Figure 1: Mean percent cover of native and non-native grasses along the Mesa (GL) transects at the North Campus Open Space (NCOS) from 2021 to 2025. Filled circles represent the grand mean across transects and open circles represent individual transect means. Shaded ribbons show  $\pm 1$  standard error. Dashed vertical lines indicate the prescribed burn (Fall 2023) and rotational sheep grazing (Fall 2024) treatment events.

## Plot 2: Sweet and Bur Clover Over Time

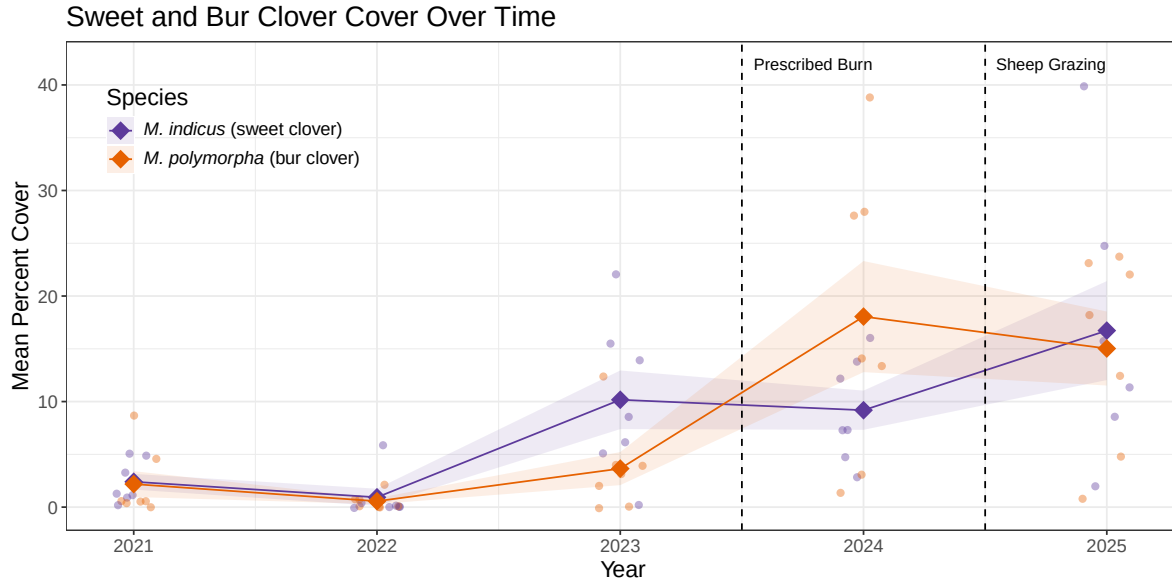
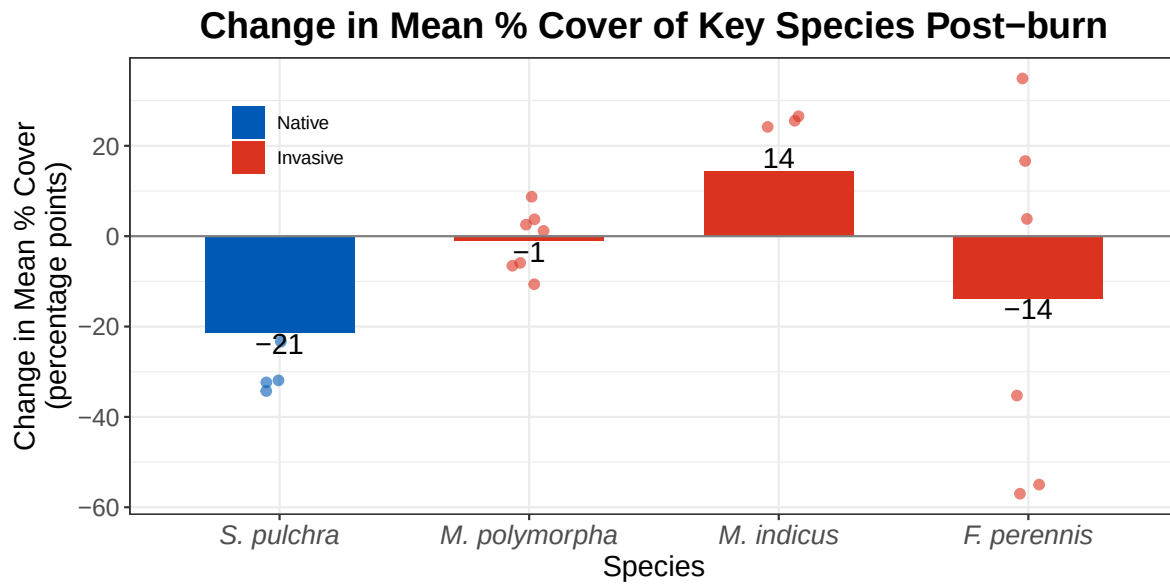
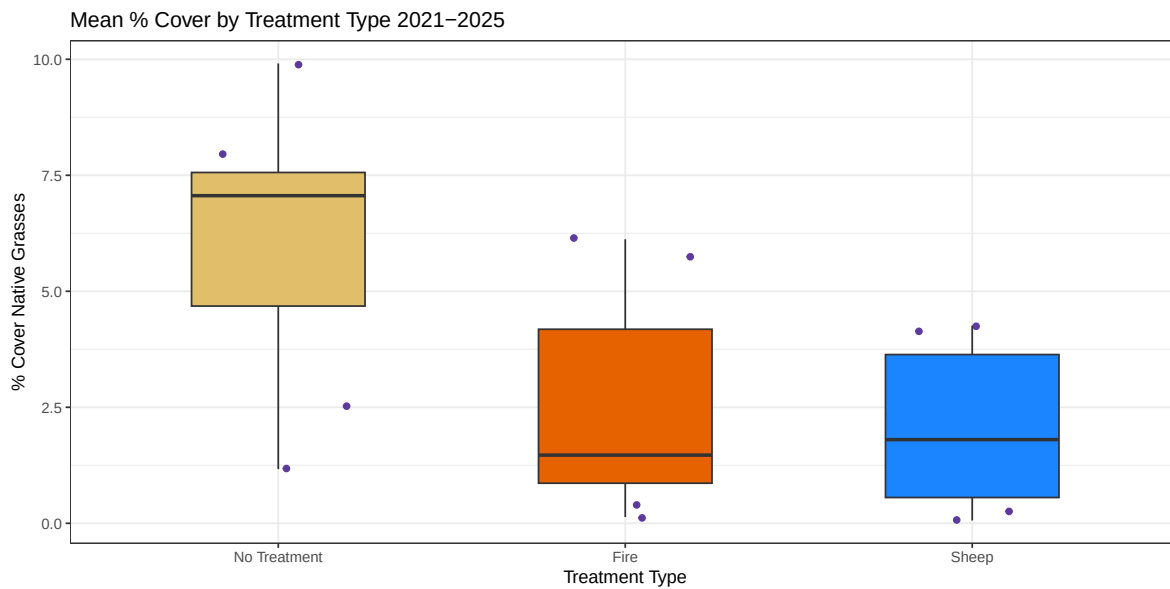


Figure 2: Mean percent cover of bur clover (*M. polymorpha*) and sweet clover (*M. indicus*) along the Mesa (GL) transects at the North Campus Open Space (NCOS) from 2021 to 2025. Filled circles represent the grand mean across transects and open circles represent individual transect means. Shaded ribbons show  $\pm 1$  standard error. Dashed vertical lines indicate the prescribed burn (Fall 2023) and rotational sheep grazing (Fall 2024) treatment events. Field observations suggested *M. polymorpha* increased following the prescribed burn and *M. indicus* following sheep grazing, a pattern reflected in the monitoring data.

**Plot 3: Change in Mean Percent Cover of Key Species Diverging Bar Chart**



**Plot 4: Disturbance boxplot**



## **Exploratory Plot 1**

## **Exploratory Plot 2**

## **Citations**

### **1. Evaluating Sheep Grazing as a Strategy for Restoring the North Campus Open Space (NCOS) Grasslands (eScholarship Cheadle Center Poster)**

[Link to paper #1](#)

Notes:

- Confirmed focal species selection; our four key species align with those identified in the poster
- Inspired figure 3 percent change bar chart from their visualization approach
- Adopted paired t-test framework comparing pre and post treatment mean percent cover
- Study is site-specific to NCOS making it directly applicable to our analysis

### **2. North Campus Open Space Restoration Project Monitoring Report: Year 7, December 2024**

[Link to paper #2](#)

Notes:

- Used for prescribed burn timing and treatment details
- Provided site context for the Mesa (GL) perennial grassland transects
- Primary data source for vegetation monitoring records used in analysis
- Covers 2017-2024 monitoring period providing pre-treatment baseline context for our 2021-2025 analysis window
- Confirms NCOS restoration scope including upland native grassland habitats relevant to our mesa transects

### **3. Trajectories of vegetation-based indicators used to assess wetland restoration progress**

[Link to paper #3](#)

Notes:

- Non-monotonic restoration trajectories – supports native cover declining post-burn not meaning restoration is failing
- Justifies need for longer-term monitoring to assess native grass recovery
- Note wetland vs. grassland limitation when citing

#### **4. A decade-long study of repeated prescription burning in California native grassland restoration**

[Link to paper #4](#)

Notes:

- Directly supports prescribed burn as a restoration tool in *Stipa pulchra* dominated California grasslands; same species and system as our study
- Contextualize the post-burn decline in our native grass cover; the paper shows *S. pulchra* recovery takes multiple years, consistent with the drop we see in 2024
- Non-native annual grasses reduced post-burn in their study, which partially aligns with our non-native grass decline post-burn
- Support longer-term monitoring; recovery patterns only became clear over 5+ years, reinforcing that our 2021-2025 window may not capture full recovery
- Note that their study used repeated burns on the same plots while ours is a single burn event; acknowledge this difference when citing

#### **5. Evaluating Prescribed Fire Effect on Medusa Head and Other Invasive Plants in Coastal Prairie at Point Pinole**

[Link to paper #5](#)

Notes:

- Coastal prairie California grassland setting makes this more ecologically comparable to NCOS mesa than most burn studies
- *Stipa pulchra* and *Brachypodium distachyon* also appear in our study providing direct species-level comparability
- Supports prescribed burn as a management tool for invasive grasses in California coastal grasslands specifically
- Note timing difference; their burn was conducted in June while ours was Fall 2023, which could affect results since burn timing influences which species are most impacted
- Methodology is comparable; transect-based vegetation monitoring pre and post burn

#### **6. Prescribed Burn Impacts on California Coastal Prairie Seed Banks**

[Link to paper #6](#)

Notes:

- Central Coast California coastal prairie setting is directly comparable to NCOS mesa; strongest geographic match of any paper so far

- Supports the finding that single prescribed burns are insufficient to reduce non-native grass abundance; contextualizes why we still see high non-native cover post-burn in our data
- Fall 2024 sampling timeline overlaps directly with our post-burn monitoring period
- Contextualizes the mixed burn effects we see; non-native grasses not declining as expected is consistent with their findings
- The point that non-native plants colonize disturbed landscapes faster than natives supports the post-burn increase we see in clover species
- Note that their study focuses on seed banks while ours measures above-ground percent cover; complementary approaches to the same question
- Useful for introducing prescribed burn management context in our background/introduction section

## **7. Grazing and Fire Management for Native Perennial Grass Restoration in California Grasslands**

[Link to paper #7](#)

Notes:

- Directly addresses combined grazing and prescribed fire as restoration tools for native perennial grass restoration in California grasslands; most directly relevant to our previous dual treatment design
- Cite to support the use of sheep grazing to reduce invasive annual grass seed banks and thatch, directly applicable to NCOS grazing treatment rationale
- *Stipa pulchra* is a focal species in this paper; supports our use of it as a native grass indicator
- Notes that perennial grasses show reduced seed production the first year post-fire but recover by year two or three; contextualizes our 2024 post-burn *S. pulchra* decline as an expected short-term response
- Recommends burning every third or fourth year since non-native annual grass composition returns to pre-fire status in about three years; useful for discussing management implications
- Seeding of annual clover noted as potentially compatible with established perennials but competitive effects on native seedlings unknown; directly relevant to our clover increase post-treatment
- Cite in methods/discussion to justify the combined burn and grazing treatment approach used at NCOS